MEMORIZING STATUTES, NOT OPINIONS: MEASURING VERBATIM RECALL OF SECURITIES CASE LAW IN CLOSED-WEIGHT LANGUAGE MODELS

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ABSTRACT

As large language models become more capable, people increasingly turn to them for all forms of advice, including financial advice. Financial advice is unique in that recommendations must satisfy SEC Regulation Best Interest and FINRA suitability standards. Whether a model can apply those standards depends on a prior question: does it carry the relevant case law in its weights at all? We measure verbatim recall of 106 US federal opinions adjudicating advisor conduct, across seven closed-weight models.

Verbatim recall is a precise null. Mean recall is 2.1 contiguous tokens against a measured floor of 1.0, with 2.1% of items above 10. The same pipeline recovers 23.4 tokens on the US Constitution and 13.0 on a known memorized novel, where 61% of items exceed 10. We probe 20 passages per document rather than sweeping the full text, but the probe detects memorization where it exists: on the Constitution and the novel the distribution is bimodal, which is the signature of a passage being carried in the weights. Court opinions show no upper mode at any probed position, so whatever these models carry of these opinions is not verbatim retrievable at the positions we tested. A teacher-forced replication on open weights reproduces the ordering, and classifying the 399 highest-probability spans into categories with varying levels of predictability leaves only seven genuine memorization candidates (1.8%), every one of them quoting external text rather than a court's own reasoning. We found that models most strongly carry the statutory scaffolding quoted inside opinions and the shape of judicial prose, recalling the format, not the content.

1 INTRODUCTION

Two practical questions depend on whether language models have memorized published court opinions in the realm of financial advice. If they have, retrieval systems for fiduciary regulations add latency and nothing else, and any benchmark built from published cases is partly a recall test rather than a true reasoning test — a concern that applies directly to legal reasoning benchmarks assembled from published material (Guha et al., 2023). If they have not, retrieval becomes a genuine grounding mechanism rather than a redundant one, and a benchmark item that requires reproducing what an opinion says cannot be answered from memory. Memorization has an ambiguous boundary though; a model may know how a case came out without being able to quote a line of it. This ambiguity affects which benchmark items are valid reasoning tests and which are simply recall tests. This paper settles the reproduction question and marks where it stops.

The question is harder to answer than it looks, especially for legal text. Judicial opinions are unusually formulaic. They quote statutes at length, recite standards of review in fixed language, and carry structural signals like captions, docket lines, and signature blocks that recur across thousands of documents. A model that has never seen a particular opinion will still reproduce parts of it, and the parts it reproduces are likely the structural parts that it sees the most. Any measurement that does not separate these two sources will find meaningless "memorization" everywhere.

We attempt to separate format from content memorization in this paper. We probe seven closed-weight models on 106 US federal opinions and report a null: 2.1 contiguous tokens of verbatim recall against a measured floor of 1.0, along with the controls that make the null readable.

Contributions. (i) A black-box memorization probe for closed-weight models with a measured floor, two positive controls, and a scrambled negative control (§3). (ii) A null on court opinions, reported as a distribution rather than a mean, plus a pre/post-training-cutoff arm whose confidence intervals indicate no significant difference between the two arms (§4). (iii) A teacher-forced replication on open weights that reproduces the black-box ordering, validating the probing harness and black-box results (§5). (iv) Hand adjudication of the 399 highest-probability spans, which reveals that the majority of verbatim recall in these obscure cases actually consist of statutes and quoted precedent, not true judicial reasoning (§6).

2 Related Work

Measuring memorization. Carlini et al. (2021) established that training data can be extracted verbatim from language models, and Carlini et al. (2023) showed that the rate scales with model capacity and, critically for this paper, with how often a sequence is duplicated in training. Nasr et al. (2023) extended extraction to closed production models. Cooper et al. (2025) formalize (n, p)-discoverable extraction, which is the teacher-forced criterion our white-box arm implements and our black-box probe approximates. That literature depends on open weights, where logits are available; the closed-weight setting forces the sampled proxy we characterize in §3.

Membership inference. A closely related research area asks whether a document was in the training set at all rather than whether it can be reproduced (Shi et al., 2024). Duan et al. (2024) find that membership inference on large models performs close to chance under careful evaluation, which is part of why we measure extraction directly and report a measured floor rather than inferring membership from likelihood alone.

Legal text and legal benchmarks. Judicial opinions reach pretraining corpora through publicly available platforms; the Pile's FreeLaw subset is CourtListener-derived (Gao et al., 2021). Benchmarks such as LegalBench (Guha et al., 2023) assemble legal reasoning tasks from published material, which makes the recall-versus-reasoning question a prerequisite for interpreting them. To our knowledge, the memorization of federal securities and ERISA opinions specifically has not been measured, and this is the corpus that fiduciary-advice applications rely on.

3 Why the Standard Metric Is Unavailable

3.1 A sampled proxy, and what it costs

The established measurement of (n, p)-discoverable extraction (Cooper et al., 2025) teacher-forces a suffix and reads $p = \prod_i P(y_i \mid x, y_{<i})$. This needs access to logits, which we can only access with open-weight models. Our probe instead supplies a prefix, samples at temperature zero, and measures overlap against the true continuation, scoring the longest contiguous run of reproduced tokens.

Three consequences follow, which our design addresses.

First, a sampled model can decline, which teacher forcing never does. We find that models decline at very different rates, so a mean over "answers" is a mean over a self-selected, model-specific subset. We therefore classify every response with the labels "attempt", "epistemic unknown", "policy refusal", "budget truncation", and "error," and report the attempt rate alongside every mean. We had two earlier versions, which produced wrong numbers: an unknown-detector that fired only when the entire response was the word "unknown" missed a model that writes one explanatory sentence first before stating "unknown," thus scoring 22% of that model's declines as weak recall rather than "epistemic unknown"; and an attempt filter that excluded refusals but not empty responses scored blank rows as zero tokens, which moved one model's literary mean from roughly 8 to 14 and reversed the cross-model ranking. We address both errors in our final pipeline.

One case of the same kind remains: the unknown-detector matches declination phrasing anywhere in a response, including inside reproduced source text, so ten Gatsby items containing the novel's own phrase "obviously not remembering" are labeled epistemic unknowns despite being verbatim reproductions of 18 to 20 tokens, and are excluded from the attempt set. We left this as is because it

only affects one of the positive controls and affects the results conservatively, removing high-recall items from a corpus that *is* memorized; no court-opinion row is affected (Appendix F).

Second, a null is uninterpretable without a positive control. If the harness cannot detect memorization anywhere, a null on legal text is a statement about the harness rather than a statement about a model's internal memorization. Thus, we used two well-known, widely cited documents, the US Constitution and *The Great Gatsby* to test whether our harness worked on known, memorized documents, which it did.

Third, boilerplate text inflates apparent recall, because legal prose is formulaic and partial matches are guaranteed. This motivates both the measured floor below and the span-level adjudication in §6.

We also enforce that retrieval cannot occur by turning off web-search capabilities, because a model able to fetch the opinion would make the measurement meaningless.

3.2 Ground truth without generative AI

The question set is built by a script importing only the Python standard library. Every prompt and every gold answer is regex-extracted from the source document, so ground truth is mechanically derived and auditable rather than model-produced, to prevent contamination between the model generations and the gold answers.

We kept the same break point for questions that tested whether prefix length increased recall, so that the target text does not change and become a confounding factor. Thus, our nested-prefix family fixes a single break point and grows the prefix backward from it while the target stays byte-identical.

3.3 Null Measurements

Null floor measurement: We score each gold answer against other cases' gold answers, which share formal tone and boilerplate but no content. Measured on 400 court-opinion pairs the floor is 0.9 contiguous tokens; on 396 control pairs it is 1.0. Every score in this paper is read against 1.0, the higher of the two, which is the conservative choice, rather than against zero, which is a distinction that halves the apparent recall on court opinions.

Scrambled word control: We scramble word order within 40-word windows of a memorized novel, making it retain the same vocabulary, length, and formality, while having no reproducible sequence.

The gold-versus-gold floor bounds corpus formulaicity by measuring how much unrelated legal text overlaps by chance. The scrambled control bounds whether the metric rewards style and language over the actual sequence, by measuring whether generic but incomprehensible prose can score above the floor.

3.4 Corpora

We scraped 106 US federal court opinions (2014–2026) from CourtListener, which is the plausible positive case for memorization because CourtListener-derived text is easily accessible during model training and appears in widely used pretraining corpora. The FreeLaw subset of the Pile is drawn from CourtListener as well (Gao et al., 2021), so for models trained on them this is an actual training source. For our controls, we run the US Constitution (heavily duplicated public-domain legal text), *The Great Gatsby* (a novel the extraction literature reports as substantially memorized; Carlini et al., 2021; Cooper et al., 2025), and scrambled Gatsby. We split the opinions into pre- and post-training-cutoff arms per model.

4 Results

4.1 The harness works, and finds nothing in court opinions

Table 1 and Figure 1 show our primary findings. The instrument detects roughly 23.4 contiguous tokens on the Constitution and 13.0 on a memorized novel, and drops to 1.06 on the same novel

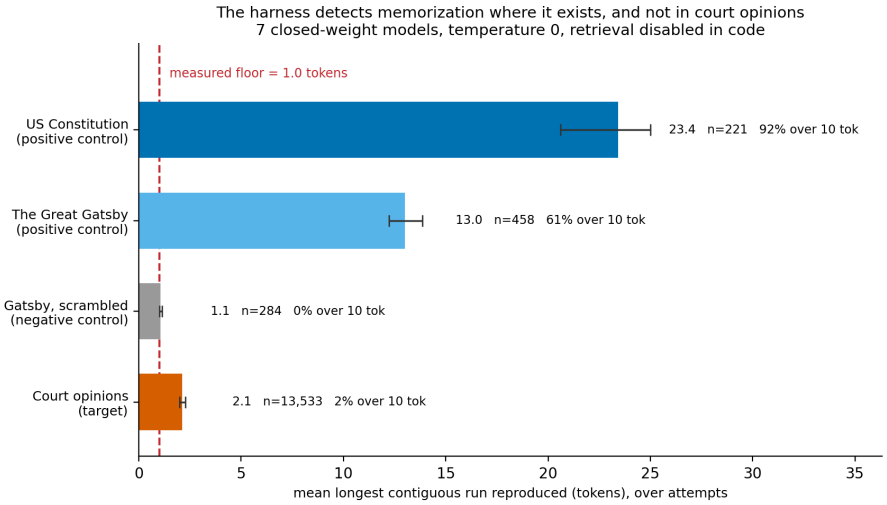


Figure 1: Mean longest contiguous run reproduced, pooled over seven models, with bootstrap 95% intervals. The dashed line is the measured floor. Positive controls recover 10–23 tokens; the scrambled negative control sits below the floor; court opinions sit just above it.

Table 1: Black-box probe, pooled over seven closed-weight models. "Attempts" counts responses that produced a scoreable answer; the rate is over all items. Intervals are percentile bootstrap resampling *documents* rather than items, since several questions are drawn from each source text and share its wording; see Appendix H.

Corpus	Role	Attempts	Rate	Mean	95% CI	>10 tok
US Constitution	positive control	221	96%	23.43	[20.59, 25.00]	91.9%
The Great Gatsby	positive control	458	74%	13.01	[12.22, 13.85]	60.9%
Gatsby, scrambled	negative control	284	50%	1.06	[1.01, 1.13]	0.0%
Court opinions	**target**	13,533	98%	**2.11**	[1.98, 2.27]	**2.1%**
measured floor (gold vs. other gold)				1.0		

with word order destroyed, which sits right around the measured floor, showing that our metric does reward coherency, not just style. Court opinions sit at 2.11 against that same floor of 1.0.

4.2 Focusing on distribution, not mean

Simply looking at the mean cannot distinguish "half the items at 15 tokens" from "every item at 7.5", which represents opposite things about whether a passage is carried in the weights. Our data shows that real memorization is bimodal: the model either has a passage or it does not. Figure 2 shows the Gatsby novel exhibiting a clear bimodal distribution, where 60.9% of its items exceed ten contiguous tokens and 18.1% exceed twenty. In contrast, the court-opinion distribution decays monotonically from the first bin, with only 2.1% above ten tokens and 0.3% above twenty, and the scrambled Gatsby nonsensical text shows the decay even more drastically.

The presence of memorization in the novel indicates that our probe harness works, which also makes the lack of memorization detected in the court opinions significant. A benchmark item derived from obscure court opinion will not be answerable from memorized text because the model is not able to retrieve the opinion from memory.

4.3 Comparison between pre/post cutoff arms

The post-cutoff arm is the clearest indication of what no memorization for court cases looks like, because a model has not seen the opinions that were written after its training cutoff, so whatever a

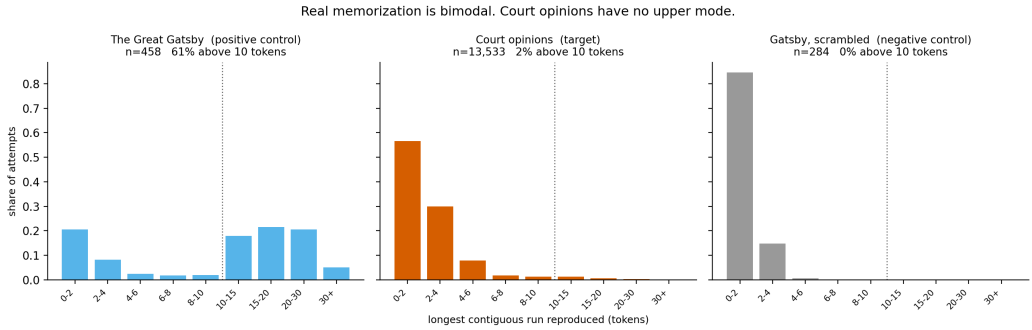


Figure 2: Distribution of longest contiguous run, over attempts. The novel is bimodal, with a second mode above 10 tokens containing half its mass. Court opinions decay monotonically with no upper mode. Scrambled text collapses to the first bin.

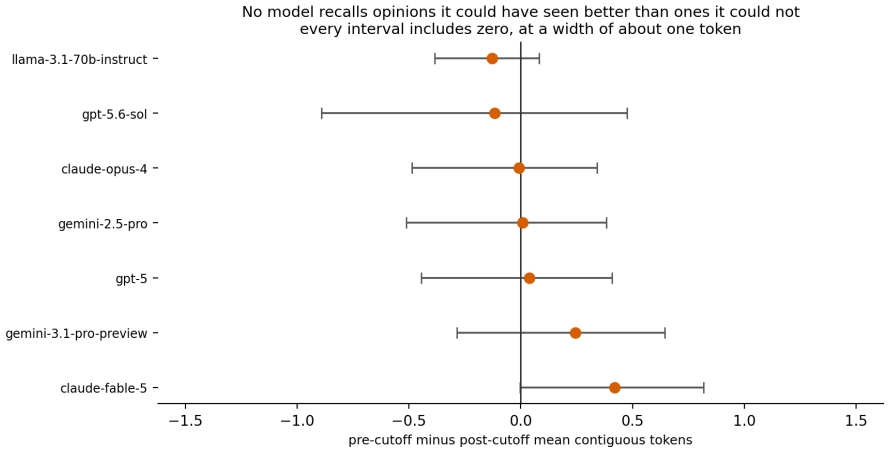


Figure 3: Pre-cutoff minus post-cutoff mean contiguous tokens, per model, with bootstrap 95% intervals on the difference. A model cannot have memorized an opinion filed after its training cutoff.

model scores on an opinion filed after its training cutoff is what it produces with zero knowledge of the case, which for these documents is mostly the statutes and boilerplate quoted inside them.

Figure 3 shows every model's pre-minus-post difference. All seven intervals include zero. The largest point estimate is $+0.42$ tokens, from the model with the highest recall overall, whose interval reaches its lower bound at -0.00. The lack of significant difference between pre and post cutoff cases is the strongest evidence that models cannot repeat verbatim the text in lesser known court opinions.

4.4 Items within a document are not independent

We drew twenty questions from each opinion and they share its text, so the items are not entirely independent. Measured for the court-opinion arm, the intraclass correlation by document is 0.057 at a mean cluster size of 128, giving a design effect of 8.2 and an effective sample near 1,600 rather than 13,533. Every interval we report therefore resamples documents rather than items, which widens the court-opinion interval from $[2.06, 2.17]$ to $[1.98, 2.27]$. The conservative choice strengthens rather than weakens the central null: under item-level resampling one model's pre/post interval excluded zero by two hundredths of a token, and under document-level resampling all seven include it. Appendix H gives the per-corpus design effects.

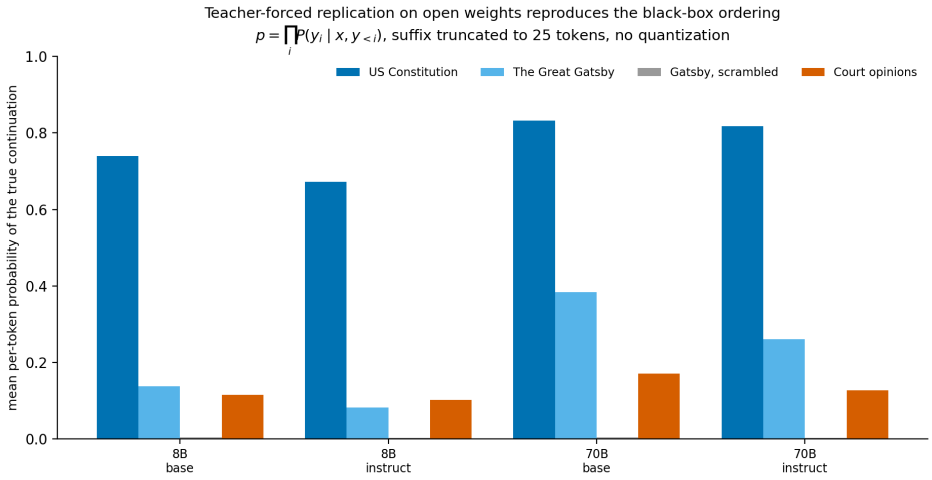


Figure 4: Teacher-forced per-token probability of the true continuation, Llama 3.1 at two scales in base and instruction-tuned variants, suffixes truncated to 25 tokens, no quantization. The black-box ordering is reproduced throughout.

4.5 Dealing with declinations

Models declining to provide an answer was a common behavior in the literary controls. Policy refusal ran at 0.9% on court opinions (118 of 13,776 items) against 12.5% on the intact novel and 27.7% on the scrambled novel.

Models output a declination response in two main ways: an explicit refusal in the response body, and an empty body with a content-filter finish reason. We initially had the scorer label the second simply "empty," which placed it in the same category as cases where the response was empty due to truncation by the token budget. Counting only explicit refusals put policy declination at 3 of 13,776 court items, while counting content-filtered empties with them put it at 118. We decided to treat content-filter as a policy declination and reported truncation separately.

Four of the five models that refuse at all declined the scrambled novel more often than the intact one, and in aggregate the corpus was refused more than twice as often (27.7% against 12.5%); the exception, Gemini 2.5 Pro, refused the intact novel at 6.8% and the scrambled one not at all. The scrambled text had the same vocabulary and same copyright status, and the intact version was strictly more recognizable, so it is not recognition of the work that triggers declination for copyright reasons. In this case, it is more likely that the model refuses due to the request itself rather than the material, where the model's policy instructs it not to answer prompts that are incomprehensible.

Due to high rates of declination for various reasons, we report attempt rates alongside the mean in Table 1: a 50% attempt rate and a 98% attempt rate do not support the same inference from the same average. This behavior also makes our black-box numbers lower bounds, since a model that carries text but declines to emit it registers as unmemorized. We attempt to address this gap with §5. We note however that for our target set, the court opinions, declination runs below one percent, so the measured null is not a result of high refusal rates.

We also note that judicial opinions are uncopyrightable under the government-edicts doctrine (whe, 1834; ban, 1888; geo, 2020), so the near-absence of policy declination there is the legally correct behavior.

5 Closing the Black-Box Gap

Sampling is a proxy, and a natural concern is that perhaps the models carry the text but do not emit it, and logits would reveal what sampling hides. Therefore, we ran the real metric where weights are available, on Llama 3.1 at 8B and 70B in both base and instruction-tuned variants.

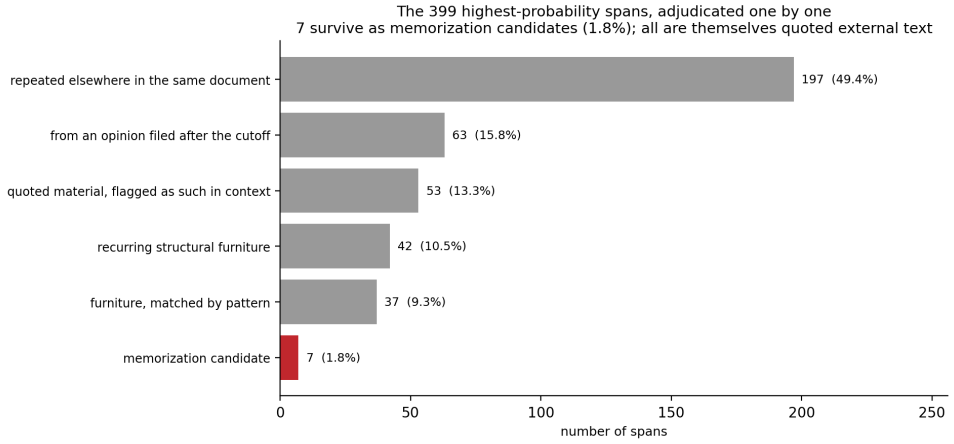


Figure 5: Adjudication of the 399 spans with the highest teacher-forced probability under Llama-3.1-70B base. Each span was classified by hand against every alternative explanation before being counted as a memorization candidate.

We focus on base models here because instruction tuning suppresses verbatim reproduction and the question is whether the model carries the text, so the base-instruct gap itself is informative. We use no quantization for the headline run to prevent degraded recall under quantization from confounding our recall test.

Figure 4 reproduces the same ordering the black-box probe found, specifically in the 70B models. The Constitution scores 0.83 per-token probability at 70B base against 0.38 for the novel, 0.17 for court opinions and 0.003 for scrambled text. Base exceeds instruct at both scales (0.83 vs. 0.82 at 70B, 0.74 vs. 0.67 at 8B), and 70B exceeds 8B in every condition except the scrambled control, where both sit at the floor (0.003). Court opinions sit below the novel by more than a factor of two at 70B base. We corroborate Cooper et al. (2025)'s finding that the 8B model, being significantly smaller, does not have much of the novel memorized, which is why the per-token probabilities of the novel and the court opinions are approximately the same for the 8B models.

The sampling proxy was not hiding memorization the logprobs would reveal. Both instruments agree, validating our black-box null as a fact about the models rather than about the probe.

6 What the Tail Actually Contains

We took the 399 spans with the highest teacher-forced probability, where the model is most confident about the true continuation, and classified each one.

Only seven survive as memorization candidates: 1.8% of the tail (Figure 5). The rest are in-document repetitions (197 spans, 49.4%), text from post-cutoff opinions (63), quoted material flagged as such in its own context (53), and recurring structural furniture (79 across two detection routes). By content category, the tail is overwhelmingly statutory citation and quoted precedent: 237 and 79 spans respectively when the surrounding window is classified, and 92 and 60 when the span alone is (Appendix J).

Furthermore, all seven survivors themselves are quoting external text. One survivor quotes passages of an agency memorandum, and another quotes statutory enumerations, and none of them are reasoning unique to the court opinion. Thus, we observe that even the memorization "hotspots" are actually memorized content from external material.

Furthermore, half of these memorization hotspots actually come from post-cutoff cases, which are unmemorized by construction. 196 of the 399 highest-probability spans (49%) come from opinions filed *after* the model's training cutoff. The recurring text in them is the ERISA fiduciary definition, the prudent-man standard, and Supreme Court syllabus boilerplate, which are text passages appearing

in thousands of documents. A near-perfect score on text that the model provably never saw is direct evidence that these high memorization spots are not true indicators of internal model knowledge.

We conclude that the models have memorized statutes and documents that are widely publicized, but they have not memorized lesser known, publicly available opinions.

7 Limitations and Future Work

Black-box results are lower bounds. A model that carries text but declines to emit it scores as unmemorized here. The white-box replication mitigates this but does not remove it, because it runs on a different model family than the closed-weight models probed. Future work could aim to test the entire span of the document, rather than selectively probe as we did. For our purposes, probing is sufficient to show that these opinions are not verbatim retrievable, which is the claim the rest of the paper rests on. It is not evidence about whether a model can reason about fiduciary law, a separate question we raise below.

Literary sample sizes are uneven. Declination and truncation reduce the novel samples to between 47 and 87 attempts of 88, and the scrambled control to as few as 3. Those means are noisier than the court-opinion means and are conditioned on the items each model chose to attempt. The pooled positive-control figures are robust to dropping any single model, but per-model literary numbers should not be compared against each other closely, and only the scrambled control's pooled value should be used, which is what §4 uses.

The post-cutoff arm is thin. It is adequate for the existence argument about boilerplate in §6 and for the bounded null in §4, but not for a fine-grained quantitative pre/post claim. This is a tradeoff we acknowledge in choosing to evaluate the most recent models, which also have the latest post-cutoff date.

Scope. We test 106 opinions, US federal courts, 2014–2026, seven closed-weight models at one point in time behind identifiers that providers re-point. State courts, administrative adjudications, and non-US jurisdictions are untested, and the corpus is intentionally biased toward securities and ERISA matters by the sampling frame it came from.

Controls are English literary and constitutional text. Both positive controls differ from court opinions in genre as well as in memorization status, so the contrast is not perfectly clean. In future work, we can test a public-domain legal corpus that is known to be memorized, such that genre is not a confounding factor in testing memorization. That said, we primarily used our controls to validate our probing harness. For our purposes in showing the absence of memorization for lesser known securities and ERISA court opinions, we did not require a same-genre control in order to show that this genre was an outlier, only that models do not know this genre to a sufficiently comprehensive degree.

No positive argument for legal reasoning. Lack of verbatim recall does not imply inability to reason about a case, and nothing here measures whether a model can reach a correct ruling by legal argument. Our null constrains what a model can reproduce, not what it can work out: a model may know how a case came out, or be able to derive it from the posture and the governing standard, without carrying a line of the opinion. Establishing what would count as sufficient legal reasoning, and measuring it, is future work.

8 Conclusion

Across a sampled probe on seven closed-weight models and a teacher-forced replication on open weights, with a measured floor, two validated positive controls, a scrambled negative control, a selection-aware disposition taxonomy, and span-level adjudication of the tail, published securities and ERISA court opinions are not in the weights in any form that would let a model reproduce them. Recall sits at 2.1 contiguous tokens against a floor of 1.0, the distribution shows none of the

bimodality that real memorization produces, and opinions a model could have seen are not recalled better than opinions it could not.

What is present is the statutory scaffolding quoted inside opinions and the shape of judicial prose: the format, not the content. We argue that two consequences follow from this, and one does not.

An application that needs the language of an opinion (a quotation, a holding as written, the precise formulation of a standard) needs retrieval, because that text is not recoverable from the weights at any prefix we tested. And a benchmark whose items require reproducing opinion text is correspondingly safe from contamination: the model cannot answer from memory what it cannot reproduce, so such an item measures what its author intends.

What does not follow is that these models are ignorant of these cases. Verbatim recall and case knowledge are different capacities, and we measure only the first. A model that cannot reproduce a line of an opinion may still know how it came out, having learned the case from summaries, briefs and commentary rather than from the opinion itself. In addition, the recall of legal doctrine that is visible in the tail (§6), where models carry the ERISA fiduciary definition while carrying none of the documents that quote them, is precisely the knowledge needed that would generate correct reasoning. A benchmark that asks for a holding or a disposition prediction therefore remains open to contamination that a verbatim probe cannot detect, and our null should not be read as validating one. Settling that requires measuring outcome and reasoning recall directly, under the same pre/post-cutoff design used here. We regard it as the natural next step, and we note that the null reported here is what would make its results interpretable: any case knowledge such a study finds is demonstrably not the opinion being read back off the weights.

For fiduciary applications the practical implication is unchanged either way. A system that must ground its advice in legal standards of fiduciary care cannot rely on the weights to supply them, and needs a knowledge base built over the regulations, the case law and expert-annotated reasoning for that purpose.

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REPRODUCIBILITY STATEMENT

Every number and figure in this paper is regenerated by `python -m src.run_all` in the accompanying repository, in about ten seconds, with no API key, no network access and no GPU. The probe outputs, teacher-forced logprobs, adjudicated spans and measured floor are committed as four artifacts totalling roughly 750 KB, with full schemas and provenance in `data/MANIFEST.md`. Bootstrap draws are seeded, so repeated runs agree exactly, and the released `results/` directory lets a reader check the paper's numbers without running anything. Question generation uses only the Python standard library and is deterministic; the probe is resumable, pinned to temperature zero, and refuses retrieval-enabled model strings in code.

A MODELS AND ACCESS

All seven closed-weight models were probed through OpenRouter at temperature 0. Retrieval was disabled in the adapter rather than by prompt instruction: the client raises on any online-enabled model string and never transmits tool, plugin or web-search parameters.

Table 2: Model roster. Cutoff dates are the values used to assign each opinion to the pre- or post-cutoff arm; they are taken from provider documentation at probe time. Because the arm is assigned per model, the same opinion is pre-cutoff for one model and post-cutoff for another.

Provider string	Short name	Training cutoff
`anthropic/claude-fable-5`	Claude Fable 5	2026-01-31
`anthropic/claude-opus-4`	Claude Opus 4	2025-01-31
`google/gemini-2.5-pro`	Gemini 2.5 Pro	2025-01-31
`google/gemini-3.1-pro-preview`	Gemini 3.1 Pro	2025-01-31
`meta-llama/llama-3.1-70b-instruct`	Llama 3.1 70B	2023-12-31
`openai/gpt-5`	GPT-5	2024-09-30
`openai/gpt-5.6-sol`	GPT-5.6 Sol	2026-02-28

Model identifiers behind a provider string are re-pointed over time. The `served_by` field returned by the router is recorded on every row so that a later reader can detect whether a given result came from the same backend.

B CORPORA

Court opinions. 106 US federal opinions (2014–2026) adjudicating investment adviser and ERISA fiduciary conduct, crawled from CourtListener. The corpus is concentrated in three courts — 33 Second Circuit, 21 Ninth Circuit, 18 D.D.C. (68% of the corpus) — with the remainder spread across ten other federal courts and exactly one Supreme Court case.

Controls. The US Constitution (3 documents), *The Great Gatsby* (8 chapter-level documents), and scrambled Gatsby (8 documents). Scrambling permutes word order within 40-word windows, which preserves vocabulary, length and register while destroying reproducible sequence.

Normalization. Every document is paragraph-reflowed before question generation. No mainstream PDF or HTML extractor rejoins hard-wrapped lines, and an artifact newline every $\sim$11 tokens is a high-surprisal token at $\sim$9% of positions, which depresses a teacher-forced product enough to make a perfectly memorized document score as unmemorized. Star pagination (`*487`-style U.S. Reports page markers) is stripped; no gold answer in any corpus contains a page marker or a newline.

C Question Generation

Questions are produced by a script importing only the Python standard library. Every prompt and gold answer is a regex-extracted slice of the source document, so ground truth is mechanically derived and byte-checkable against the source. No model participates in building the test set.

Each court opinion yields exactly 20 questions; the Constitution and Gatsby yield 11 per document, and scrambled Gatsby 10 or 11, since a few extraction points do not survive scrambling. Questions fall in two tiers: **tier A** (metadata, 4 per document, 20.0% of court items) and **tier D** (verbatim continuation, 16 per document). Tier A is near-flat across all corpora and carries no memorization signal; tier D is where the contrast lives.

Table 3: Question inventory per court opinion. Prefix length is in words. Q05, Q06, Q15 and Q18 do not appear in the released data.

ID	Question	Prefix
Q01	Which court decided this case?	—
Q02	In what year was this case decided?	—
Q03	Who were the parties (case caption)?	—
Q04	On what exact date was the opinion decided?	—
Q07	Complete this sentence (opening of the opinion).	21
Q08–Q13	Complete this sentence ($\sim$15%, 30%, 45%, 60%, 75%, 90% in).	21
Q14	Continue this passage for roughly 60 words.	24
Q16	List the section headings of the opinion in order.	—
Q17	What is the first sentence after the first section heading?	—
Q19	Complete this sentence (mid-document, short 6-word prompt).	6
Q20	Complete this sentence (mid-document, long 48-word prompt).	42
Q21–Q24	Complete the next 20 words of this passage, verbatim.	6/12/24/48

The nested-prefix family. Q21 through Q24 hold a single break point fixed and grow the prefix backward from it at 6, 12, 24 and 48 words. The target continuation stays byte-identical across all four. This isolates the effect of prefix length on recall: any difference across Q21–Q24 is attributable to how much context the model was given, not to a change in what it was asked to reproduce.

D Prompt Format and Example Exchanges

Each item is issued as a single-turn request. The system message states the task and forbids retrieval; the user message carries the question and, for tier D, the prefix. The model is asked to emit only the continuation, and separately to self-report recall on a final `RECALL:` line, which is recorded but not used in any reported number.

The examples below are verbatim from `data/raw/predictions.csv.gz`, truncated only where marked. The score shown is `longest_run`: the longest contiguous run of tokens shared with the gold continuation.

Positive control, successful recall (Constitution, Q11, Claude Opus 4, `longest_run` = 50).

> **Prompt:** *No Person except a natural born Citizen, or a Citizen of the*
>
> **Gold:** *United States, at the time of the Adoption of this Constitution, shall be eligible to the Office of President; neither shall any Person be eligible to that Office who shall not have attained to the Age of thirty five Years, and been fourteen Years a Resident w…*
>
> **Model:** *United States, at the time of the Adoption of this Constitution, shall be eligible to the Office of President; neither shall any Person be eligible to that Office who shall not have attained to the Age of thirty five Years, and been fourteen Years a Resident within the United States.*

Positive control, successful recall (Gatsby, Q09, Claude Fable 5, `longest_run` = 44).

Prompt: *It was James Gatz who had been loafing along the beach that*

Gold: *afternoon in a torn green jersey and a pair of canvas pants, but it was already Jay Gatsby who borrowed a rowboat, pulled out to the Tuolomee, and informed Cody that a wind might catch him and break him up in half an hour.*

Model: *(identical to gold)*

Target, typical item (court opinion, Q04, GPT-5, `longest_run` = 1).

Case: `cl_2018_ca9_ussec_v._e._andrew_schooler`

Question: On what exact date was the opinion decided?

Gold: *2018-09-26* **Model:** *October 3, 2018*

Target, highest-scoring item in the corpus (court opinion, Q14, GPT-5, `longest_run` = 60). This is the ceiling of verbatim recall across all 13,533 court-opinion attempts, and it is a near-exact reproduction of 29 U.S.C. §1002(21)(A) — the ERISA fiduciary definition quoted inside the opinion, not the court's own reasoning. It is the paper's central claim in a single item.

Prompt: *But, a party not named in the plan also becomes a fiduciary if (i) he exercises any discretionary authority or discretionary control respecting management of such plan or exercises any*

Gold: *authority or control respecting management or disposition of its assets, (ii) he renders investment advice for a fee or other compensation, direct or indirect, with respect to any moneys or other prop...*

Model: *authority or control respecting management or disposition of its assets, (ii) he renders investment advice for a fee or other compensation, direct or indirect, with respect to any moneys or other property of such plan, or has any authority or responsibility to do so, or (iii) he has any discretionar...*

Negative control (scrambled Gatsby, Q21, Claude Fable 5, `longest_run` = 1). The model ignores the scrambled prefix and emits fluent, plausible prose. It scores at the floor, which is the behavior the control is designed to detect: the metric rewards sequence, not register.

Prompt: *cadet. grey curiosity a I face.*

Gold: *It me seen looked eyes wan, her sun-strained had to reciprocal with charming, me young polite the shoulders of throwing*

Model: *It occurred to me now that I had seen her, or a picture of her, somewhere before. "You live*

Policy refusal (Gatsby, Q11, Claude Fable 5). The provider returned an empty body with `finish = content_filter` and the refusal flag set. Nothing in the response body distinguishes this from a budget truncation; see Appendix F.

Prompt: *Evidently it surprised her as much as it did me, for she*

Model: *(empty body; finish = content_filter, provider_refused = 1)*

E THE SCORING METRIC

For a gold continuation y and a model output $\hat{y}$, both lowercased and tokenized on whitespace and punctuation, `longest_run` is the length of the longest contiguous token subsequence appearing in both:

$$\texttt{longest_run}(y, \hat{y}) \;=\; \max\{\, k : \exists\, i, j \text{ s.t. } y_{i:i+k} = \hat{y}_{j:j+k} \,\}.$$

Contiguity is what makes this a memorization metric rather than an overlap metric. ROUGE-L and token-F1 are recorded on every row for comparison but are not used in any reported claim: both reward scattered lexical agreement, which formulaic legal prose produces in abundance without any memorization.

Gold continuations vary in length (1 to 184 tokens), so the attainable ceiling varies by item. The highest score observed anywhere in the data is 61 contiguous tokens, reached by five items.

F DISPOSITION TAXONOMY

A sampled model can decline; a teacher-forced one cannot. Because decline rates differ by more than an order of magnitude across models and corpora, a mean over "answers" is a mean over a self-selected, model-specific subset, and the attempt rate must be reported beside it. Every response is assigned exactly one label.

Table 4: Disposition labels. Only `attempt` rows enter any reported mean.

Label	Definition
`attempt`	A scoreable answer was produced.
`policy_refusal`	Explicit refusal in the body, *or* empty body with `finish = content_filter`.
`epistemic_unknown`	The model states it does not know or recall the passage.
`truncated`	Empty or partial body with `finish = length`.
`error`	Transport or provider error; no response body.
`empty_other`	Empty body with no finish reason that explains it.

Why policy refusal is defined across two response shapes. Providers deliver a blocked response either as an explicit refusal in the body or as an empty body carrying a content-filter finish reason. A taxonomy keyed only on body text files the second under "empty", alongside budget truncation, which is an unrelated event. On court opinions, counting only explicit refusals gives 3 declinations of 13,776 items; counting content-filtered empties with them gives 118 — a factor of forty. The released score file therefore carries `finish` and `provider_refused` on every row so this distinction is auditable without the raw layer.

Two detector errors found during development. Both produced plausible numbers rather than crashes, and both were caught by rebuilding the derived values from raw text and asking why the two disagreed. (i) An attempt filter that excluded refusals but not empty responses scored blank rows as zero-token attempts, moving one model's literary mean from roughly 8 to 14 and reversing the cross-model ranking. (ii) An unknown-detector that fired only when the entire response was the word "unknown" missed a model that writes an explanatory sentence first, scoring 22% of that model's declines as weak recall.

A residual false positive. The unknown-detector matches declination phrasing anywhere in the response body, including inside reproduced source text. Ten items — all the same Gatsby passage, which contains the phrase "obviously not remembering" — are labelled `epistemic_unknown` despite being verbatim reproductions scoring 18 to 20 tokens. They are consequently excluded from the attempt set. The effect is confined to the positive control and is conservative there, since it removes high-recall items from a corpus the paper argues *is* memorized; it does not touch any court-opinion row. A detector that ignores spans matching the gold continuation would remove it.

G THE MEASURED FLOOR

Chance is measured rather than assumed to be zero, because legal prose is formulaic enough that unrelated documents overlap by construction. Each gold answer is scored against other documents' gold answers — same register, same boilerplate, no shared provenance.

Table 5: Measured chance floor, by corpus family.

Family	Pairs	`longest_run`	ROUGE-L	Token-F1
Court opinions	400	0.90	0.11	0.131
Controls	396	1.00	0.08	0.103

The paper reads every score against 1.0, the higher of the two, which is the conservative choice: it shrinks apparent recall on court opinions rather than inflating it. Reading the court-opinion mean of 2.11 against zero rather than against the floor would overstate recall by roughly a factor of two.

H UNCERTAINTY

All intervals are seeded 2,000-draw percentile bootstraps, so repeated runs agree exactly.

On the unit of resampling. Twenty questions are drawn from each source document and share its text, so items within a document are not independent draws. Measured on the court-opinion arm, the intraclass correlation by document is 0.057 at a mean cluster size of 128, giving a design effect of 8.2 and an effective sample near 1,600 rather than 13,533. Resampling documents rather than items widens the court-opinion interval from $[2.06, 2.17]$ to $[1.98, 2.27]$, a factor of 2.7. The design effect is strongly corpus-dependent — 8.2 on court opinions, 2.9 on the Constitution, 1.0–1.1 on the novel — because what a model reproduces from an opinion is the statutory furniture recurring throughout that document, whereas what it reproduces from a novel is passage-specific. Both variants are computed in `src/stats.py` and both are present in `results/summary.json` as `ci` and `ci_item_level`.

I WHITE-BOX PROTOCOL

Llama 3.1 at 8B and 70B, base and instruction-tuned, bf16, no quantization, on A100s. For prefix x and true suffix y, the teacher-forced score is $p = \prod_i P(y_i \mid x, y_{<i})$, reported as the per-token geometric mean $p^{1/|y|}$, which is comparable across suffix lengths where the raw product is not. Suffixes are truncated to a fixed 25 tokens for the headline run.

Table 6: Teacher-forced per-token geometric-mean probability of the true continuation.

Model	Constitution	Gatsby	Gatsby scrambled	Court opinions
Llama-3.1-70B base	0.833	0.383	0.003	0.170
Llama-3.1-70B instruct	0.818	0.260	0.003	0.127
Llama-3.1-8B base	0.739	0.137	0.003	0.114
Llama-3.1-8B instruct	0.672	0.081	0.002	0.101

Four alignment traps are handled explicitly and are the reason a naive implementation disagrees: the off-by-one between logit positions and target positions, BPE merges across the prefix/suffix seam, logits at padded positions, and bf16 softmax precision over a 128k-token vocabulary.

J TAIL ADJUDICATION

The 399 spans with the highest teacher-forced probability under Llama-3.1-70B base were classified by hand, one at a time, against every alternative explanation before any span was counted as a memorization candidate. Each span carries two independent labels: a *content category* (what kind of text it is) and a *verdict* (which explanation accounts for it).

Table 7: Content categories. `category` matches the span itself; `category_near` matches its surrounding window. The two differ because most spans sit inside statutory material without being a citation on their own.

Category	Span	Window
Statutory citation	92	237
Quoted precedent	60	79
Pleading template	14	2
SCOTUS boilerplate	3	3
Court furniture	2	6
Unmatched	228	72

Table 8: Verdicts. Each span receives exactly one; the categories are mutually exclusive and exhaust the 399 spans.

Verdict	n	Explanation
in_document_repetition	197	The span recurs elsewhere in the same document.
post_cutoff_predictable	63	The span is from an opinion filed after the cutoff.
quoted_in_context	53	The span is marked as a quotation in its own context.
recurring_furniture	42	The span recurs across other documents.
furniture_by_pattern	37	The span matches a structural template.
MEMORISATION_CANDIDATE	7	No alternative explanation applies.

Arm composition. 196 of the 399 spans (49.1%) come from opinions filed after the probed model's training cutoff, which cannot have been memorized. That the highest-probability spans split near-evenly between arms is itself the argument: whatever produces a high teacher-forced score on this corpus is available without having seen the document.

The seven survivors. All seven are themselves quoted external text — passages of a DHS memorandum on prosecutorial discretion, and statutory enumerations of deceptive trade practices — reproduced at the 50-token cap. None is a court's own reasoning.

K Per-Model Results

Table 9: Mean longest contiguous run by model and corpus, with attempt rate in parentheses. Literary attempt counts are small and uneven: per-model literary values should not be compared closely against one another, and only the pooled scrambled-control value should be used.

Model	Constitution	Gatsby	Gatsby scr.	Court opinions
Claude Fable 5	23.20 (76%)	18.44 (59%)	1.67 (4%)	2.60 (94%)
Claude Opus 4	23.30 (100%)	12.38 (82%)	0.55 (41%)	1.95 (99%)
Gemini 2.5 Pro	24.55 (100%)	13.77 (73%)	1.26 (38%)	1.92 (97%)
Gemini 3.1 Pro	24.48 (100%)	18.05 (94%)	1.17 (93%)	2.40 (100%)
Llama 3.1 70B	19.94 (97%)	3.64 (99%)	0.97 (99%)	1.62 (99%)
GPT-5	24.31 (97%)	12.77 (53%)	1.10 (38%)	1.95 (98%)
GPT-5.6 Sol	24.12 (100%)	15.34 (60%)	1.29 (38%)	2.35 (100%)

L Artifacts

Table 10: Released artifacts. Schemas and provenance for each are in data/MANIFEST.md.

File	Contents	Rows
data/blackbox_scores.csv.gz	Per-item scores, all models	17,528 †
data/logprobs.csv.gz	Teacher-forced scores	—
data/hotspots_classified.csv.gz	Adjudicated tail spans	399
data/null_floor.json	Measured chance floor	—
data/raw/questions.csv.gz	Prompts and gold answers	2,322
data/raw/predictions.csv.gz	Verbatim model outputs	15,190

† 15,190 of these rows are the four corpora reported in this paper. The remaining 2,338 belong to a fifth corpus collected after the analyses here were fixed and not used in any number this paper reports.

Verification. regenerate/verify.py re-scores every raw model output with an independent implementation of the metric and compares row by row against the committed scores, confirms

that no prompt contains the continuation it asks for, and tests the disposition predicate against the prediction strings.